

APJ ABDUL KALAM TECHNOLOGICAL UNIVERSITY

Fifth Semester B.Tech Degree Examination December 2021 (2019 scheme)

Course Code: HUT310**Course Name: MANAGEMENT FOR ENGINEERS**

Max. Marks: 100

Duration: 3 Hours

PART A*(Answer all questions; each question carries 3 marks)*

Marks

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|----|---|-----|
| 1 | What are the different levels of management? | (3) |
| 2 | What is scientific management? | (3) |
| 3 | Explain the importance of delegation in management. | (3) |
| 4 | “Planning is looking ahead and control is looking back” Comment | (3) |
| 5 | How can we improve the productivity of an organisation? | (3) |
| 6 | List out the steps in rational decision-making. | (3) |
| 7 | Differentiate between CPM and PERT. | (3) |
| 8 | Explain the dummy activity in a project network. | (3) |
| 9 | Differentiate between needs, wants, and demands. | (3) |
| 10 | Define entrepreneurship. | (3) |

PART B*(Answer one full question from each module, each question carries 14 marks)***Module -1**

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|----|---|------|
| 11 | a) Explain the systems approach to management with a block diagram. | (10) |
| | b) How are changes in the external environment compensated in the systems approach? | (4) |
| 12 | a) Explain Task and Responsibilities of a professional Manager. | (8) |
| | b) Explain the importance of management in an organisation. | (6) |

Module -2

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|----|---|-----|
| 13 | a) Explain the mission, goals, strategy, programmes and procedures of an organisation with suitable examples. | (8) |
| | b) Differentiate between strategic and tactical decisions. | (6) |
| 14 | a) Illustrate the different types of organisation structures. | (8) |
| | b) Explain the factors governing the selection of organisation structures. | (6) |

Module -3

- 15 a) The producer of an apple crates company produces 270 crates per 100 logs with his current equipment. He currently purchases 100 logs per day, and each log required 3 labour hours to process. He believes that he can hire a professional buyer who can buy a better quality log at the same cost. If this is the case, he increases his production to 290 crates per 100 logs. His labour hours will increase by 8 hours per day. What will be the impact on productivity (measured in crates per labour-hour) if the buyer is hired? What is the growth in productivity in this case? (10)
- b) Explain decision making under uncertainty. (4)
- 16 a) A cell phone manufacturer has invented a 3D phone. The company wants to take decision whether to manufacture the phone, take royalty from another manufacturer, or sell rights of the invention and take a lump sum amount of ₹50,000. The profits associated and probability of these alternatives is given in the table below. Represent the problem as a decision tree and suggest a decision to maximise profits. (7)

Demand	Manufacture the phone		Take royalty	
	Probability	Profit (₹)	Probability	Profit (₹)
High	0.25	200,000	0.25	60,000
Medium	0.4	50,000	0.4	40,000
Low	0.35	-10,000	0.35	20,000

- b) Explain the different models of decision-making behaviour. (7)

Module -4

- 17 a) A project consists of 7 activities and the three time estimates are as follows: (10)

Activities	Duration in weeks		
	t_o	t_m	t_p
A	2	6	10
B	4	6	12
C	2	3	4
D	2	4	6
E	3	6	9
F	6	10	14
G	1	3	5

The sequence of activities is as follows:

- Activities *A* and *B* start at the beginning of the project.
- When *A* is completed *C* and *D* start.
- *E* can start when *B* and *D* are finished.

- F can start when B , C and D are completed and is the final activity.
 - G can start when F is finished and is the final activity.
 - i. What is the expected time of the duration of the project?
 - ii. Identify the critical path of the project.
 - iii. Calculate the probability that project will be completed in 23 weeks.
- b) Write notes on Fulkerson's rule of numbering events. (4)
- 18 a) The following table shows the precedence requirements, normal and crash times, and normal and crash costs for a project. The indirect costs are ₹ 70/day. (10)

Activity	Required Time (days)		Cost (₹)	
	Normal	Crash	Normal	Crash
1-2	8	6	100	200
1-3	4	2	150	350
2-4	2	1	50	90
2-5	10	5	100	400
3-4	5	1	100	200
4-5	3	1	80	100

- i. Draw the network.
 - ii. Determine the critical path.
 - iii. Determine the optimal duration and the associated cost after crashing.
- b) Differentiate between activity and event in a project network. (4)
- Module -5**
- 19 a) What are intellectual property rights? Explain the business importance of patents. (7)
- b) Explain the various interrelationships between the following functional areas. (7)
- (i) Production and Marketing
 - (ii) Production and Finance
 - (iii) Production and Personnel.
- 20 a) Explain the process of market segmentation. How is the marketing mix related to market segmentation? (10)
- b) "Operations management is process of planning, organizing and controlling the activities of a production function". Explain. (4)
